

Interference-Aware Routing: Evaluation Report

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1 Experiment Setup

Simulator: ncsim, HEFT scheduler, `csma_bianchi` 802.11ax interference model. **RF config:** 5 GHz, 20 MHz channel, $P_{\text{tx}} = 20$ dBm, path-loss exponent $n = 3.0$. **Grid spacing:** 40 m (adjacent links ≈ 8.6 MB/s PHY; diagonals ≈ 56.6 m, lower PHY rate). **Seeds:** Each configuration averaged over 30 seeds (smoothing HEFT non-determinism from Python hash randomization). **Workload:** Heterogeneous compute costs (150–1000) and data sizes (2–30 MB).

1.1 Networks

Network	Nodes	Links	Topology
4×4 grid	16	68	Grid + checkerboard diagonals, heterogeneous compute
7×7 grid	49	240	Grid + checkerboard diagonals, heterogeneous compute

1.2 DAGs

DAG	Tasks	Edges	Structure
Small	8	12	Fork-join: 1 source $\rightarrow$ 6 parallel $\rightarrow$ 1 sink
Large (4×4)	30	48	5-stage pipeline with cross-connections
Large (7×7)	60	102	6-stage pipeline with cross-connections

Label	Scheme	Description
W	Widest-Path	Max bottleneck bandwidth; ignores interference
S	Shortest-Path	Min total latency ($\equiv$ min hops at equal latency); ignores interference
GS	Greedy-Start	Static greedy at DAG inject; flows sorted by est. start time
GC	Greedy-Criticality	Static greedy; flows sorted by HEFT upward rank (desc.)
GB	Greedy-Bytes	Static greedy; flows sorted by data size (desc.)
GO	Greedy-Overlap	Static greedy; flows sorted by overlap degree (desc.)
GSD	Dynamic	Routes computed at <code>transfer_start</code> using actual active-link state
GSD-D	Dynamic + Deferral	Like GSD, but defers transfers when congestion exceeds threshold

Experiment	W	S	GS	GC	GB	GO	GSD	GSD-D	Best
4×4 , small	191.1	20.4	50.9	50.9	50.9	50.9	76.0	48.3	S
4×4 , large	363.5	234.1	183.8	203.4	178.4	193.6	298.7	158.4	GSD-D
7×7 , small	250.9	35.8	187.8	184.6	202.4	192.8	230.2	199.2	S
7×7 , large	4852.9	2465.1	3644.7	3850.1	3518.1	3873.5	3603.5	3629.5	S

1.3 Schemes Evaluated

2 Results

2.1 Mean Makespan (seconds, averaged over 30 seeds)

2.2 Improvement over Widest-Path Baseline

2.3 Winners

3 Analysis

3.1 Shortest-Path Dominates with Heterogeneous Workloads

S wins 3 of 4 experiments. With heterogeneous data sizes (2–30 MB), min-hop paths are especially valuable for large transfers. A 30 MB transfer on a 1-hop path avoids the multi-hop relay chains that trigger severe `csma_bianchi` interference. S’s advantage is largest on small DAGs (–86–89% vs W) where contention is low enough that simply avoiding unnecessary hops is optimal.

3.2 GSD-D Wins on 4×4 with Large DAGs

On the 4×4 grid with 30 tasks, GSD-D (158.4s) is the clear winner, beating GB (178.4s) by 11% and S (234.1s) by 32%. This is the **only scenario where deferral helps**: the small, dense grid forces many transfers onto the same few links, and deferral allows transfers to wait for less-congested conditions.

Notably, GSD-D dramatically outperforms its non-deferral counterpart GSD (298.7s vs 158.4s, –47%). Without deferral, GSD’s per-flow greedy optimization makes locally optimal but globally

Experiment	S	GS	GC	GB	GO	GSD	GSD-D
4×4 , small	−89%	−73%	−73%	−73%	−73%	−60%	−75%
4×4 , large	−36%	−49%	−44%	−51%	−47%	−18%	−56%
7×7 , small	−86%	−25%	−26%	−19%	−23%	−8%	−21%
7×7 , large	−49%	−25%	−21%	−28%	−20%	−26%	−25%

Experiment	Winner	Runner-up	Gap
4×4 , small	S (20.4 s)	GSD-D (48.3 s)	57.7%
4×4 , large	GSD-D (158.4 s)	GB (178.4 s)	11.2%
7×7 , small	S (35.8 s)	GC (184.6 s)	80.6%
7×7 , large	S (2465.1 s)	GB (3518.1 s)	29.9%

suboptimal choices on the dense grid. Deferral breaks this cascade by letting the engine wait for airtime to free up.

3.3 Deferral Does Not Help on 7×7 Grids

On the 7×7 grid, GSD-D (199.2 s small, 3629.5 s large) provides no improvement over the static greedy variants. The larger grid offers enough alternative paths that transfers can route around congestion without deferring. Deferral adds latency without commensurate benefit.

3.4 Greedy Orderings Diverge with Heterogeneous Data

With heterogeneous data sizes, the greedy orderings now show meaningful differentiation. On 4×4 large, GB (178.4 s) beats GS (183.8 s) by 3%—GB’s largest-first ordering gives the 30 MB transfers the best routes. On 7×7 large, GB (3518.1 s) outperforms all other greedy variants by 2–10%.

The four static greedy variants are no longer tied on 4×4 small (50.93 s each). With uniform data sizes they were identical; with heterogeneous sizes, the ordering happens to produce the same routes for this simple fork-join topology.

3.5 Widest-Path Remains Consistently the Worst

W is 5–9× worse than the best scheme. With heterogeneous data sizes, the penalty for long relay chains is even more severe: a 30 MB transfer on a 3-hop Widest-Path route experiences compounding interference on each link.

4 Regime Map

Scenario	Best approach	Why
Small grid, large DAG	GSD-D	Dense contention; deferral reduces interference
Small grid, small DAG	S	Few transfers; min-hop avoids interference
Large grid, any DAG	S	Enough paths to spread traffic; min-hop is sufficient
Heterogeneous data sizes	GB	Largest-first ordering gives big transfers best routes

5 Deferral Analysis

5.1 When Deferral Helps

GSD-D’s deferral mechanism (threshold=0.3, max_deferrals=3) is effective only on the 4×4 grid with a large DAG. This is the regime with the highest link contention: 48 data edges compete for 68 links on a 16-node grid. The deferral mechanism allows transfers to wait 1–2 transfer completions before starting, finding less-congested paths.

5.2 When Deferral Hurts

On the 7×7 grid, deferral adds no value and slightly hurts performance compared to static greedy schemes. The larger grid provides enough routing diversity that transfers can always find an acceptable path immediately. Deferring only delays the transfer without improving the path quality on retry.

5.3 Comparison: GSD vs GSD-D

Experiment	GSD	GSD-D	Improvement
4×4 , small	76.0	48.3	−36%
4×4 , large	298.7	158.4	−47%
7×7 , small	230.2	199.2	−13%
7×7 , large	3603.5	3629.5	+1%

Deferral consistently improves on GSD (−13 to −47%) except on 7×7 large where it is essentially neutral (+1%). The largest gain is on 4×4 large (−47%), confirming that deferral is most valuable in high-contention regimes.

6 Reproducing These Results

```
cd ncsim/  
python run_routing_eval.py  
# Output: /tmp/ncsim_routing_eval/eval_results.json  
# Runtime: ~50 minutes (960 runs)
```